Alligation:

Live Class Campusmonk TCS / Other Batch

Pre-assessment:

Question1: If the average weight of the whole class is 50 kg. and the average weight of boys in the class is 60 kg and the average weight of girls in the same class is 45kg. What could be the possible strength of boys and girls in the class respectively? **TCS NQT**

A) 20, 30 (B) 30, 20 (C) 10, 5 (D) 10, 20 (E) 10, 30

Question2: A person bought 2 varieties of tea, 1st at Rs.40/kg and 2nd at Rs.70/kg. He mixed both varieties and sold the mixture at Rs.55/kg making a profit of 10%. In what ratio did he mix the two varieties of tea?

(A) 1:2 (B) 2:3 (C) 3:2 (D) 2:1

Type: Concept



Question1: In what proportion must a grocer mix rice at Rs.12 a kg and Rs.7 a kg so as to make a mixture worth Rs. 8 a kg?

7:12
1:4
2:3
12:7

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Question2: In what proportion must tea at Rs 47.50 a kg be mixed with tea at Rs 50.50 per kg to produce a mixture worth Rs 48.50 per kg?

(A) 2:1 (B) 1:2 (C) 4:1 (D) 3:2

The ratio in which two sugar solutions of the concentration 15% and 40% are to be mixed to get a solution of concentration 30% is:

2:3
3:2
8:9
9:8

Question3: In what ratio must 25% of alcohol be mixed with 50% of alcohol to get a mixture of 40% strength alcohol?

Question 4: Two vessels A and B contain mixtures of milk and water. The ratio of milk and water in these two vessels is 5:2 and 8:5 respectively. In what ratio must these mixtures be mixed to form a new mixture containing milk and water in 9:4 ratios? 1) 7:2 2) 2:7 3) 3:5 4) 5:3 5) 7:9

Question5: Two vessels A and B contain milk and water mixed in the ratio 4: 3 and 2: 3. In what ratio must these mixtures be mixed to form a new mixture containing half milk and half water?

1) 7: 5 2) 1: 2 3) 2: 1 4) 6: 5 5) None of these

Question6: In what ratio should two different types of mixtures containing, milk and water in the ratio of 5:1 and 2:1 respectively mixed to obtain a final mixture containing milk and water in the ratio of 3:1?

1) 7:5 2) 1:2 3) 2:1 4) 1:1 5) None of these

Question7: Two vessels A and B contain spirit and water mixed in the ratio 5:2 and 7:6 respectively. Find the ratio in which this mixture be mixed to obtain a new mixture in vessel c containing spirit and water in the ratio 8:5?

A.1:7 B.2:9 C.7:9 D.3:8

Question8 : Two vessels A and B contain milk and water mixed in the ratio 4:3 and 2:3. The ratio in which these mixtures be mixed to form a new mixture containing half milk and half water is:

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(A) 7:5

(B) 6:5

(C) 5:6

(D) 4:3

Type:

Question1: What quantity of wheat at rate Rs. 7.4 per kg mixed with 35 kg of wheat which rate Rs. 6.4 per kg so that price of mixed wheat become Rs. 6.7 per kg??

(A) 12kg (B) 15kg (C) 21kg (D) 10kg

Question2: A shopkeeper bought one type of rice at Rs 12 per kg and other type at Rs 16.25 per kg. After mixing both types of rice he fixed the cost of mixture as Rs 14.75 per kg. If the total quantity of the rice be 85kg, find the quantity of first type of rice.

(A) 55Kg (B) 30Kg (C) 35Kg (D) 40Kg

Question3: Two types of rice x Rs/kg and 50 Rs/kg, 2kg of first type mixed with 6 kg of second type than rate of mixed rice becomes 45 Rs/kg. Find the value of x?

(A) 20 (B) 25 (C) 60 (D) 30

Type: Average

Question1:

If the average weight of the whole class is 50 kg. and the average weight of boys in the class is 60 kg and the average weight of girls in the same class is 45kg. What could be the possible strength of boys and girls in the class respectively?

A) 20, 30 (B) 30, 20 (C) 10, 5 (D) 10, 20 (E) 10, 30

Question2:

In a class of 100 students, the average weight is 30 kg. If the average weight of the girls is

24 kg and that of the boys is 32 kg, then what is the number of girls in the class?

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Question3:

A company has undergraduate and graduate employee. The average salary of each graduate employee is Rs 16550 and undergraduate employee is Rs 11950, average salary of workers in company is Rs 14750 if there are total 575 workers in company then

find out total number of graduate employees in company?
(A) 280 (B) 350 (C) 301 (D) 322

Type : Mixtures

Question1: A mixture of 40 litres of milk and water contains 10% water. How much water must be added to take 20% water in the new mixture?

(A) 5 litres (B) 6 litres (C) 8 litres (D) 10 litres

Question2: 300 grams of sugar solution has 40% of sugar in it. How much sugar should be added to make it 50% in the solution?

A) 10 gms B) 60 gms C) 40 gms D) 80 gms

Question3: 150 gram of sugar solution has 20% sugar in it. How much sugar should be added to make it 25% in the solution ?

Question4: In 50gm alloy of gold and silver. The gold is 80% by weight. How much gold should be mixed to this alloy so that the weight of gold would become 95%?

Question5: A mixture of 40 litres of milk and water contains 10% of water. How much water is to be added to the mixture so that the water may be 20% in the new mixture.

Type: Profit & Loss

Question1: A merchant has 840 kg of rice, part of which he sells at 9% profit and the rest at 16% profit. He gains 12% profit on the whole. What is the quantity sold at 16% profit?
1) 480 kg 2) 240 kg 3) 360 kg 4) 600 kg 5) None of these

Question 2: A trader has 4000 kg of tea, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. What quantity of tea was sold at 8% profit?
1) 480 kg 2) 200 kg 3)2400 kg 4) 1600 kg 5) None of these

Question 3: A person bought 2 varieties of tea, 1st at Rs.40/kg and 2nd at Rs.70/kg. He mixed both varieties and sold the mixture at Rs.55/kg making a profit of 10%. In what ratio did he mixed the two varieties of tea?
(B) 1:2 (B) 2:3 (C) 3:2 (D) 2:1

Question4: How much of salt at 42 paise/kg must be mixed with 25 kg of salt at 24 paise/kg, so that a profit of 25% is obtained by selling the mixture at 40 paise/kg?
1) 25 kg 2) 20 kg 3) 30 kg 4) 24 kg 5) None of these